Graph-theory exercises

1. Prove that every simple graph has an even number of odd-degree vertices.
2. Decide whether (4,3,2,2,1) is a simple graph's degree sequence. If so draw one; otherwise prove impossibility with a graphical criterion or the handshake lemma.
3. A connected eight-vertex graph has exactly two degree-three vertices and all others degree four. Find its edge count and decide existence of Euler trails and circuits.
4. Prove $K_{3,3}$ nonplanar without relying on a drawing's visible crossings.
5. Derive the chromatic polynomial of C_4 using deletion–contraction or a classification of colorings.
6. Prove a bridge cannot belong to a Hamilton cycle. Join two K_5 graphs by one bridge. Which vertices fail Dirac's condition? Explain why the ten-vertex graph has a Hamilton path but no Hamilton cycle.
7. For the text's two K_4 drawings, list the common vertex and edge sets and explain why the crossing does not add a vertex. In the plane drawing delete AD, BD, AC in order; give v, e, f and prove each deleted edge is currently a nonbridge. For disjoint K_3 and K_2, compute v, e, f, c and explain why outer faces cannot be counted separately.
8. A digraph has arcs AB, BC, CA, AC, CD, DA, where letter order gives direction. List indegrees/outdegrees, determine Euler circuit/open trail existence, and give a full legal edge order. Add a disconnected directed triangle XYZ. What happens to degrees and why does the prior conclusion no longer follow?
9. A graph has vertices $(i, j) \in \mathbb{Z}_6 \times \mathbb{Z}_7$, adjacent when exactly one coordinate changes by ± 1 modulo its own modulus. Find vertex count, degree, edge count; rigorously prove triangle-freeness and use a planar bound to prove nonplanarity. Drawing crossings do not suffice.
10. On five fixed labels, count simple graphs with exactly four edges, without isolates, and connected. Show intermediate formula terms and explain the difference between the last two counts.
11. Prove a Hamilton path implies $c(G-S) \leq |S|+1$ for every vertex set S. Use this for $K_{2,4}$, then decide Hamilton path and cycle existence in $K_{3,4}$.
12. Derive chromatic polynomials of C_3,C_4,C_5 by deletion–contraction and compute $P_{C_5}(2)$,$P_{C_5}(3)$. Explain their relation to odd cycles and proper colorings. Do not quote a general cycle formula.

Graph-theory exercises (continued)

13. For edges e1=AB, e2=BC, e3=CA, e4=CD, e5=DE, e6=EC, run the stack algorithm from A, scanning each adjacency list in increasing edge-ID order. Record S and answer after every push and pop. Why does the first pop not end the search? Give the final vertex and edge-ID sequences. Next replace EC by EF, where F is a new vertex: determine the only permissible start vertices for an open Euler trail, and explain what fails if the same code starts at A without checking its preconditions.
14. Fix triangle ABC with leaf D drawn inside and CD as its only incident edge. Edge IDs 1..4 denote AB, BC, CA, CD. Give both boundary walks and lengths, all dual edge endpoints, a dual drawing, its vertex degrees and face count. Independently delete AB or CD from the original, keeping every vertex, and check $v - e + f = 1 + c$. If D and CD are instead drawn outside the triangle, where does the dual loop attach? Which total counts remain unchanged?

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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